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COMMON QUARTERLY EXAMINATION - 2026 Reg. No.

XII - MATHEMATICS

Time Allowed : 3.00 Hrs. *Maximum Marks: 90*

Part - I

I. Choose the correct answer. **20 x 1 = 20**

1. $A=\begin{bmatrix}2 & 0\\ 1 & 5\end{bmatrix}$ and $B=\begin{bmatrix}1 & 4\\ 2 & 0\end{bmatrix}$ then $|\text{adj}(AB)|=$

   a) - 40 b) - 80 c) - 60 d) - 20

2. If $(AB)^{-1}=\begin{bmatrix}12 & -17\\ -19 & 27\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}1 & -1\\ -2 & 3\end{bmatrix}$ then $B^{-1}=$

   a) $\begin{bmatrix}2 & -5\\ -3 & 8\end{bmatrix}$ b) $\begin{bmatrix}8 & 5\\ 3 & 2\end{bmatrix}$ c) $\begin{bmatrix}3 & 1\\ 2 & 1\end{bmatrix}$ d) $\begin{bmatrix}8 & -5\\ -3 & 2\end{bmatrix}$

3. If $A=\begin{bmatrix}\frac{3}{5} & \frac{4}{5}\\ x & \frac{3}{5}\end{bmatrix}$ and $A^T = A^{-1}$, then the value of x

   a) $\frac{-4}{5}$ b) $\frac{-3}{5}$ c) $\frac{3}{5}$ d) $\frac{4}{5}$

4. The area of the triangle formed by complex numbers z, iz and z + iz in the Argand's diagram is

   a) $\frac{1}{2}|z|^2$ b) $|z|^2$ c) $\frac{3}{2}|z|^2$ d) $2|z|^2$

5. The solution of the equation $|z| - z = 1 + 2i$ is

   a) $\frac{3}{2} - 2i$ b) $-\frac{3}{2} + 2i$ c) $2 - \frac{3}{2}i$ d) $2 + \frac{3}{2}i$

6. If $(1+i)(1+2i)(1+3i)\ldots\ldots(1+ni) = x + iy$, then $2.5.10\ldots\ldots(1+n^2)$ is

   a) 1 b) i c) $x^2 + y^2$ d) $1 + n^2$

7. If α, β and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum\frac{1}{\alpha}$ is

   a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$

8. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is

   a) 2 b) 4 c) 1 d) ∞

9. If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$, then $\cos^{-1} x + \cos^{-1} y$ is equal to

   a) $\frac{2\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π

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10. The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is

a) [1,2] b) [-1,1] c) [0,1] d) [-1,0]

11. $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$, then the value of x is

a) 4 b) 5 c) 2 d) 3

12. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

13. If P (x,y) be any point on $16x^2 + 25y^2 = 400$ with foci F_1 (3,0) and F_2 (-3, 0) then $PF_1 + PF_2$ is

a) 8 b) 6 c) 10 d) 12

14. The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is

a) $\frac{\sqrt{3}}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$

15. If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to

a) 2 b) -1 c) 1 d) 0

16. If $\vec{a}(\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$, where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{b}.\vec{c} \neq 0$ and $\vec{a}.\vec{b} \neq 0$ then $\vec{a}$ and $\vec{c}$ are

a) perpendicular b) parallel

c) inclined at an angle $\frac{\pi}{3}$ d) inclined at an angle $\frac{\pi}{6}$

17. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then

a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < c < 1$

18. If ρ represents the rank and, A and B are n x n matrices, then

a) $\rho(A+B) = \rho(A) + \rho(B)$ b) $\rho(AB) = \rho(A).\ \rho(B)$

c) $\rho(A-B) = \rho(A) - \rho(B)$ d) $\rho(A+B) \leq n$

19. The domain of $\operatorname{cosec}^{-1} x$ function is

a) $\mathbb{R} \setminus (-1,1)$ b) $\mathbb{R} \setminus \{-1,1\}$ c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ d) $\mathbb{R} - \{0\}$

20. For the parabola $(x-h)^2 = -4a(y-k)$, the equation of the directrix is

a) $y = k$ b) $y = a$ c) $x = k + a$ d) $y = k + a$

Part - II

II. Answer any 7 questions. (Q.No.30 is compulsory) **7 x 2 = 14**

21. If $\text{adj}(A) = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$, find A^{-1}.

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22. If $\omega \neq 1$ is a cube root of unity, show that $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}+\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2}=-1$.

23. Find the modulus and principal argument of complex number $\sqrt{3}+i$

24. Find a polynomial equation of minimum degree with rational coefficients, having $2i+3$ as a root.

25. Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$.

26. Find the general equation of the circle whose diameter is the line segment joining the points (-4,-2) and (1,1).

27. Identify the type of conic sections of the equation $11x^2-25y^2-44x+50y-256=0$

28. Find the volume of the parallelopiped whose coterminous edges are represented by the vectors $-6\hat{i}+14\hat{j}+10\hat{k}$, $14\hat{i}-10\hat{j}-6\hat{k}$ and $2\hat{i}+4\hat{j}-2\hat{k}$.

29. Prove that $[\vec{a}-\vec{b},\ \vec{b}-\vec{c},\ \vec{c}-\vec{a}]=0$.

30. Show that the distance from the origin to the plane $3x+6y+2z+7=0$ is 1

Part - III

III. Answer any 7 questions. (Q.No.40 is compulsory) **7 x 3 = 21**

31. Find the rank of the matrix $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$ by reducing it to a row-echelon form.

32. Solve the following system of linear equations by matrix inversion method

    $2x - y = 8,\ 3x + 2y = -2$

33. Find the values of the real numbers x and y, if the complex numbers $(3 - i)x - (2 - i)y + 2i + 5$ and $2x + (-1 + 2i)y + 3 + 2i$ are equal

34. Find the square roots of $6 - 8i$

35. Show that the equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solutions.

36. Find the value of $\tan^{-1}(-1)+\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$.

37. Find the angle between the straight line $\vec{r}=(2\hat{i}+3\hat{j}+\hat{k})+t(\hat{i}-\hat{j}+\hat{k})$ and the plane

    $2x - y + z = 5$.

38. A particle acted on by constant forces $8\hat{i}+2\hat{j}-6\hat{k}$ and $6\hat{i}+2\hat{j}-2\hat{k}$ is displaced from the point (1,2,3) to the point (5,4,1). Find athe total work done by the forces.

39. Find the Centre and radius of the circle $3x^2+(a+1)y^2+6x-9y+a+4=0$

40. A concrete bridge is designed as a parabolic arch. The road over bridge is 40m long and the maximum height of the arch is 15m. Write the equation of the parabolic arch. Take (0,0) as the vertex.

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Part - IV

IV. Answer all the questions. 7 x 5 = 35

41. a) Solve the following systems of linear equations by Gaussian elimination method:
$2x - 2y + 3z = 2,\ x + 2y - z = 3,\ 3x - y + 2z = 1$ ***(OR)***

b) If $\cos\alpha + \cos\beta + \cos\gamma = \sin\alpha + \sin\beta + \sin\gamma = 0$, show that

i) $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3\cos(\alpha+\beta+\gamma)$ and

ii) $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3\sin(\alpha+\beta+\gamma)$.

42. a) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$.
(OR)

b) Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

43. a) Prove that $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$.
(OR)

b) Using vector method, prove that $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$

44. a) Investigate the values of λ and μ the system of linear equations
$2x + 3y + 5z = 9,\ 7x + 3y - 5z = 8,$
$2x + 3y + \lambda z = \mu$, have
i) no solution
ii) a unique solution
iii) an infinite number of solutions.
(OR)

b) Find the non-parametric form of vector equation and Cartesian equation of the plane passing through the point (0,1,-5) and parallel to the straight lines
$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + s(2\hat{i} + 3\hat{j} + 6\hat{k})$ and $\vec{r} = (\hat{i} + 3\hat{j} + 5\hat{k}) + t(\hat{i} + \hat{j} - \hat{k})$.

45. a) If $2 + i$ and $3 - \sqrt{2}$ are roots of the equation
$x^6 - 13x^5 + 62x^4 - 126x^3 + 65x^2 + 127x - 140 = 0$ find all roots.
(OR)

b) For the ellipse $4x^2 + y^2 + 24x - 2y + 21 = 0$, find the centre, vertices and the foci. Also prove that the length of latus rectum is 2.

46. a) Prove by vector method that the perpendiculars (altitudes) from the vertices to the opposite sides of a triangle are concurrent.
(OR)

b) Find the equation of the tangent and normal to hyperbola
$12x^2 - 9y^2 = 108$ at $\theta = \frac{\pi}{3}$

47. a) Find the domain of $\sin^{-1}(2 - 3x^2)$.
(OR)

b) On lighting a rocket cracker, it gets projected in a parabolic path and reaches a maximum height of 4m when it is 6m away from the point of projection. Finally, it reaches the ground 12m away from the starting point. Find the angle of projection.

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